

Quantum Attention Algorithm

Data:

$$q_i \in \mathcal{Q}, \quad i \in [n]$$

$$(k_j, v_j) \in K \times V, \quad j \in [m]$$

Initialization:

- learn $p: K \rightarrow V$ such that $p(k_j) \approx v_j, j \in [m]$

$$\text{- let } q'_i \triangleq \sum_{j=1}^m \gamma_{ij} v_j,$$

where $\gamma_{ij} \triangleq \text{softmax}_j \text{ score}(q_i, k_j)$

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Query update loop:

- input $q'_i \in Q'$, $i \in [n]$

- learn $m: Q \rightarrow Q'$ such that $m(q_i) =_\epsilon q'_i$

- learn $\sigma: Q \otimes V \rightarrow Q' \otimes K$ such that:

$$i) \quad \sigma \begin{bmatrix} q_i \\ p(k_j) \end{bmatrix} =_\epsilon \begin{bmatrix} m(q_i) \\ k_j \end{bmatrix},$$

$$ii) \quad \pi_1 \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix} =_\epsilon m(q_i),$$

$$iii) \quad p \pi_2 \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix} =_\epsilon v_j, \quad i \in [n], j \in [m].$$

- let $\alpha_{ij} = \text{softmax score}(q'_i, k_j)$ and $j^* = \arg \max_j \alpha_{ij}$

$$\text{- let } \vec{q}_i \triangleq \pi_1 \sigma \begin{bmatrix} q'_i \\ v_{j^*} \end{bmatrix} \in Q'$$

$$\text{- let } q''_i \triangleq \sum_{j=1}^m \beta_{ij} v_j, \quad \beta_{ij} \triangleq \text{softmax score}(\vec{q}_i, k_j)$$

- replace q'_i with q''_i , and repeat*

Output:

$$q_i'' = \sum_{j=1}^m \beta_{ij} v_j \quad i \in [n]$$

$$v_{j^*} \text{ where } j^* = \arg \max_j \beta_{ij}$$

Comments:

- Learn NN equations up to $\bar{\epsilon}$ via regularization.
- Iterate query update loop until perplexity is acceptable

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* For some semantics $p: K \rightarrow V$

Lyapunov conditions are met,

ensuring convergence of query trajectories

near equilibrium points.